Technology Stack (Architecture & Stack)

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Platform	ServiceNow (Service Catalog, Flow Designer, Business Rules)
Document	Phase 2: Requirement Analysis - Technology Stack (Architecture & Stack)

1. System Components & Technology Stack (Table 1)

S.No	Component	Description	Technology / ServiceNow Module
1	User Interface	Self-service catalog request portal for submitting vehicle requests.	ServiceNow Service Portal / Maintain Items (sc_cat_item)
2	Form & Variables	7 field inputs with choices Mini, SUV, Sedan.	Service Catalog Variable Engine (item_option_new)
3	Workflow Engine	Low-code process automation for approvals, tasks, and notifications.	Flow Designer / Workflow Studio (sys_hub_flow)
4	Business Logic Engine	Server-side script execution on database record state changes.	Business Rules / GlideRecord JavaScript API
5	Notification Engine	Automated email notification generation for requesters.	ServiceNow System Policy Email Log / Send Email Action

6	Database Storage	Relational database tables storing REQ, RITM, sc_task, and incident records.	ServiceNow MariaDB / GlideRecord (sc_request, sc_req_item, sc_task, incident)
7	Deployment Package	Version management and code migration package.	ServiceNow Update Sets (sys_update_set)

2. Application Characteristics (Table 2)

S.No	Characteristics	Design Consideration & Implementation
1	Low-Code Platform Standard	Leverages standard ServiceNow Workflow Studio without modifying core instance scripts.
2	Security & Access Controls	Secured using ServiceNow System ACLs, User Roles (catalog_admin, transport_team), and System User execution context.
3	Scalable Architecture	Built on ServiceNow 3-Tier Enterprise Cloud Architecture (Presentation, Application Logic, Database).
4	High Availability	Multi-datacenter redundancy with automatic failover provided by ServiceNow Enterprise Cloud.
5	Auditing & Traceability	Full audit log on RITM record including Approval History journal entries, task assignment logs, and incident links.